

Eigenvalue Problems

June 22, 2025

1 Introduction

Let U be a bounded open subset of a Euclidean space, consider the elliptic boundary-value problem

$$\begin{cases} Lw = \lambda w & \text{in } U \\ w = 0 & \text{on } \partial U. \end{cases}$$

Theorem 1 (Weak maximum principle). *Assume $u \in C^2(U) \cap C(\bar{U})$ and*

$$c \equiv 0 \quad \text{in } U.$$

If

$$Lu \leq 0 \quad \text{in } U,$$

then

$$\max_{\bar{U}} u = \max_{\partial U} u.$$

Theorem 2 (Third Existence Theorem for weak solutions).

(i) *There exists at most countable set $\Sigma \subset \mathbb{R}$ such that the boundary-value problem*

$$\begin{cases} Lu = \lambda u + f & \text{in } U \\ u = 0 & \text{on } \partial U \end{cases}$$

has a unique weak solution for each $f \in L^2(U)$ if and only if $\lambda \notin \Sigma$.

(ii) *If Σ is infinite, then $\Sigma = \{\lambda_k\}_{k=1}^{\infty}$, the values of a nondecreasing sequence with*

$$\lambda_k \rightarrow +\infty$$

Remark. From this we know that the set Σ of eigenvalues of L is at most countable.

Theorem 3 (Eigenvalues of symmetric elliptic operators).

(i) *Each eigenvalue of L is real.*

(ii) *Furthermore, if we repeat each eigenvalue according to its (finite) multiplicity, we have*

$$\Sigma = \{\lambda_k\}_{k=1}^{\infty},$$

where

$$0 < \lambda_1 \leq \lambda_2 \leq \lambda_3 \leq \dots$$

and

$$\lambda_k \rightarrow \infty \quad \text{as } k \rightarrow \infty.$$

(iii) *Finally, there exists an orthonormal basis $\{w_k\}_{k=1}^{\infty}$ of $L^2(U)$, where $w_k \in H_0^1(U)$ is an eigenfunction corresponding to λ_k :*

$$\begin{cases} Lw_k = \lambda_k w_k & \text{in } U \\ w_k = 0 & \text{on } \partial U \end{cases}$$

for $k = 1, 2, \dots$

Remark.

1. Owing to the regularity theory, and furthermore $w_k \in C^\infty(\bar{U})$ if ∂U is smooth, for $k = 1, 2, \dots$
2. The principal eigenvalue λ_1 is **simple**. In particular

$$0 < \lambda_1 < \lambda_2 \leq \lambda_3 \leq \dots$$

3. We have the **Rayleigh's formula**

$$\lambda_1 = \min_{\substack{u \in H_0^1(U) \\ u \neq 0}} \frac{B[u, u]}{\|u\|_{L^2(U)}^2}$$

4. In the proof of the simplicity of principal eigenvalue, the **strong maximum principle** is used.
5. **Weyl's asymptotic formula**

$$\lim_{k \rightarrow \infty} \frac{\lambda_k^{\frac{n}{2}}}{k} = \frac{(2\pi)^n}{|U|\alpha(n)}$$

Example 1. Consider $M = [0, L]$. Then for $k = 1, 2, \dots$,

$$\begin{aligned}\phi_k(x) &= \sqrt{2/L} \sin(\pi k x / L), \\ \lambda_k &= (\pi k / L)^2\end{aligned}$$

is the solution of **Dirichlet** eigenvalue problem.

For the **Neumann** eigenvalue problem the boundary conditions are

$$\phi'(0) = \phi'(L) = 0.$$

Take $\phi_k(x) = \sqrt{2/L} \cos(\pi k x / L)$, from which one has

$$\lambda_k = (\pi(k-1)/L)^2,$$

and the Weyl formula is again satisfied.

In the case of the **mixed** eigenvalue problem one has boundary conditions

$$\phi'(0) = \phi(L) = 0$$

or

$$\phi(0) = \phi'(L) = 0.$$

Take $\phi_k(x) = \sqrt{2/L} \sin(\pi(k-1/2)x/L)$, one has

$$\lambda_k = \left(\pi \left(k - \frac{1}{2} \right) / L \right)^2.$$

Weyl's formula follows immediately.

Finally, let us roll $[0, L]$ into a circle and consider a **closed** eigenvalue problem

$$\omega(x) = (L/2\pi) e^{i2\pi x/L}$$

One has

$$\begin{aligned}\lambda_1 &= 0 \\ \lambda_{2k} &= \lambda_{2k+1} = (2\pi k/L)^2\end{aligned}$$

Remark. From above we know the first non-trivial eigenvalue of a closed manifold is not simple.

2 Eigenvalue estimates

Theorem 4 (Lichnerowicz and Obata). *Let M be an m -dimensional compact manifold without boundary. Suppose that the Ricci curvature of M is bounded from below by*

$$\mathcal{R}_{ij} \geq (m-1)K$$

for some constant $K > 0$, then the first nonzero eigenvalue of the Laplacian on M must satisfy

$$\lambda_1 \geq mK$$

Moreover, equality holds if and only if M is isometric to a standard sphere of radius $1/\sqrt{K}$.

By multiplying ϕ by a constant it is possible to arrange that

$$a - 1 = \inf_M \phi, \quad a + 1 = \sup_M \phi,$$

where $0 \leq a(\phi) < 1$ is the median of ϕ .

Lemma 1 (Li-Yau). *Suppose M^m is a compact manifold without boundary and its Ricci curvature is nonnegative. Then the first nontrivial eigenvalue satisfies*

$$\lambda_1 \geq \frac{\pi^2}{(1+a)d^2},$$

where d is the diameter of M .

Theorem 5 (Zhong-Yang). *Suppose M is a compact manifold without boundary whose Ricci curvature is nonnegative. Let $a \geq 0$ be the median of a normalized first eigenfunction with $a + 1 = \sup \phi$ and $a - 1 = \inf \phi$, and let d be the diameter. Then the first nontrivial eigenvalue satisfies*

$$d^2 \lambda_1 \geq \pi^2 + \frac{6}{\pi} \left(\frac{\pi}{2} - 1 \right)^4 a^2 \geq \pi^2 (1 + 0.02a^2).$$

Remark. From the example of circle, we get the sharp estimate

$$\lambda_1 \geq \frac{\pi^2}{d^2}. \quad (1)$$

Theorem 6 (Li-Yau). *Let M be a compact m -dimensional Riemannian manifold without boundary. Suppose that the Ricci curvature of M is bounded from below by*

$$\mathcal{R}_{ij} \geq -(m-1)R$$

for some constant $R \geq 0$, and d denotes the diameter of M . Then there exist constants $C_1(m), C_2(m) > 0$ depending on m alone, such that the first nonzero eigenvalue of M satisfies

$$\lambda_1 \geq \frac{C_1}{d^2} \exp(-C_2 d \sqrt{R}). \quad (2)$$

Corollary 1 (Li-Yau). *Let M be a compact m -dimensional Riemannian manifold whose boundary is convex in the sense that the second fundamental form is nonnegative with respect to the outward pointing normal vector. Suppose that the Ricci curvature of M is bounded from below by*

$$\mathcal{R}_{ij} \geq -(m-1)R$$

for some constant $R \geq 0$, and d denotes the diameter of M . Then there exist constants $C_1(m), C_2(m) > 0$ depending on m alone, such that the first nonzero Neumann eigenvalue of M satisfies

$$\lambda_1 \geq \frac{C_1}{d^2} \exp(-C_2 d \sqrt{R}). \quad (3)$$

Remark. The quantities $\lambda_1 d^2$ and $d\sqrt{R}$ are both invariant under scaling.

Theorem 7 (Li-Schoen). *Let M be a complete manifold of dimension m . Let $p \in M$ be a fixed point such that $B_p(2\rho) \cap \partial M = \emptyset$ for $2\rho \leq d$. Assume that the Ricci curvature on $B_p(2\rho)$ satisfies*

$$\mathcal{R}_{ij} \geq -(m-1)R$$

for some constant $R \geq 0$. For any $\alpha \geq 1$, there exist constants $C_1(\alpha)$, $C_2(m, \alpha) > 0$, such that for any compactly supported function f on $B_p(\rho)$

$$\int_{B_p(\rho)} |\nabla f|^\alpha \geq C_1 \rho^{-\alpha} \exp\left(-C_2(1 + \rho\sqrt{R})\right) \int_{B_p(\rho)} |f|^\alpha.$$

In particular, the first Dirichlet eigenvalue of $B_p(\rho)$ satisfies

$$\mu_1 \geq C_1 \rho^{-2} \exp\left(-C_2(1 + \rho\sqrt{R})\right). \quad (4)$$

Theorem 8 ([2] Cheng, the eigenvalue comparison theorem). *Let M be a compact Riemannian manifold of dimension m and ∂M be the boundary of M . Assume that the Ricci curvature of M is bounded by*

$$\mathcal{R}_{ij} \geq (m-1)K$$

for some constant K . Let us consider $\bar{M}$ to be the simply connected space form with constant curvature K . Let $\mu_1(M)$ be the first nonzero eigenvalue of the Dirichlet Laplacian on M and i be the inscribe radius of M . If $\bar{B}(i)$ is a geodesic ball in $\bar{M}$ with radius i and $\mu_1(\bar{B}(i))$ is its first Dirichlet eigenvalue, then

$$\mu_1(M) \leq \mu_1(\bar{B}(i)).$$

When $\partial M = \emptyset$, let $\lambda_1(M)$ be the first nonzero eigenvalue of the Laplacian and d be the diameter of M . If $\bar{B}(d/2)$ is a geodesic ball in $\bar{M}$ with radius $d/2$ and $\mu_1(\bar{B}(d/2))$ is its first Dirichlet eigenvalue, then

$$\lambda_1(M) \leq \mu_1\left(\bar{B}\left(\frac{d}{2}\right)\right).$$

Theorem 9. *Let M^m be a complete Riemannian manifold of dimension m . Assume that the geodesic ball $B_p(2\rho) \cap \partial M = \emptyset$. Suppose that the Ricci curvature on $B_p(2\rho)$ is bounded from below by*

$$\mathcal{R}_{ij} \geq -(m-1)R$$

for some constant $R \geq 0$. If u is a positive function defined on $B_p(2\rho) \subset M$ satisfying

$$\Delta u = -\lambda u$$

for some constant $\lambda \geq 0$, then there exists a constant C depending on m such that

$$\frac{|\nabla u|^2}{u^2}(x) \leq \frac{(4(m-1)^2 + 2\epsilon) R}{4 - 2\epsilon} + C((1 + \epsilon^{-1}) \rho^{-2} + \lambda)$$

for all $x \in B_p(\rho)$ and for any $\epsilon < 2$. Moreover, if $\partial M = \emptyset$ with $\mathcal{R}_{ij} \geq -(m-1)R$ everywhere and u is defined on M , then

$$\frac{|\nabla u|^2}{u^2}(x) \leq \frac{(m-1)^2 R}{2} - \lambda + \sqrt{\frac{(m-1)^4 R^2}{4} - (m-1)^2 \lambda R}$$

and

$$\lambda \leq \frac{(m-1)^2 R}{4}. \quad (5)$$

Definition 1. Let M^m be a complete Riemannian manifold without boundary. We denote the greatest lower bound for the L^2 -spectrum of the Laplacian Δ by $\mu_1(M)$.

The notation $\mu_1(M)$ does not necessarily mean that it is an eigenvalue of Δ , but is motivated by the characterization of $\mu_1(M)$ by

$$\mu_1(M) = \lim_{i \rightarrow \infty} \mu_1(D_i)$$

for any compact exhaustion $\{D_i\}$ of M .

Corollary 2 (Cheng). *Let M^m be a complete Riemannian manifold without boundary. Suppose the Ricci curvature of M is bounded from below by*

$$\mathcal{R}_{ij} \geq -(m-1),$$

then the greatest lower bound of the L^2 -spectrum of the Laplacian has an upper bound given by

$$\mu_1(M) \leq \frac{(m-1)^2}{4}.$$

Lemma 2 (Fischer-Colbrie-Schoen). *Let M^m be a complete Riemannian manifold, and $\mu_1(M)$ be the greatest lower bound of Δ . Then the value μ satisfies $\mu \leq \mu_1$ if and only if the operator $L = \Delta + \mu$ admits a positive solution of the equation*

$$Lu = 0$$

on M .

Remark. Caution that such a u s.t. $Lu = 0$ may not be the eigen function, since it's not necessary in $L^2(M)$! Consider the real line example.

Also recall

Theorem 10 (Yau, [7]). *Let u be a non-negative smooth subharmonic function on M^n . Then*

$$\int_M u^p = \infty \tag{6}$$

for $p > 1$, unless u is a constant function.

Example 2. Let $M^m = \mathbb{H}^m$ be the hyperbolic m -space with constant curvature -1 . Using the upper half-space model, $\mathbb{H}^m$ is given by $\mathbb{R}_+^m = \{(x_1, x_2, \dots, x_m) \mid x_m > 0\}$ with metric

$$ds^2 = x_m^{-2} (dx_1^2 + \dots + dx_m^2).$$

For $x = (x_1, \dots, x_m)$ let us consider the function $u(x) = x_m^\alpha$. Direct computation yields

$$\begin{aligned} |\nabla u|^2 &= x_m^2 \sum_{i=1}^m \left(\frac{\partial u}{\partial x_i} \right)^2 \\ &= \alpha^2 u^2 \end{aligned}$$

and

$$\begin{aligned} \Delta u &= x_m^2 \sum_{i=1}^m \frac{\partial^2 u}{\partial x_i^2} - (m-2)x_m \frac{\partial u}{\partial x_m} \\ &= \alpha(\alpha+1-m)u. \end{aligned}$$

Theorem 11 (Choi-Wang). *Let M^m be a compact, embedded, oriented minimal hypersurface in a compact, oriented Riemannian manifold N^{m+1} . Suppose that the Ricci curvature of N is bounded from below by*

$$R_{ij} \geq mR$$

for some constant $R > 0$. Then the first nonzero eigenvalue of M has a lower bound given by

$$\lambda_1(M) \geq \frac{mR}{2}. \tag{7}$$

Definition 2. If $\partial M \neq \emptyset$, we define the Dirichlet α -isoperimetric constant of M by

$$ID_\alpha(M) = \inf_{\substack{\Omega \subset M \\ \partial\Omega \cap \partial M = \emptyset}} \frac{A(\partial\Omega)}{V(\Omega)^{\frac{1}{\alpha}}}$$

where the infimum is taken over all subdomains $\Omega \subset M$ with the property that $\partial\Omega$ is a hypersurface not intersecting ∂M .

Definition 3. The Neumann α -isoperimetric constant of M is defined by

$$IN_\alpha(M) = \inf_{\substack{\partial\Omega_1 = S = \partial\Omega_2 \\ M = \Omega_1 \cup S \cup \Omega_2}} \frac{A(S)}{\min\{V(\Omega_1), V(\Omega_2)\}^{1/\alpha}},$$

where the infimum is taken over all hypersurfaces S dividing M into two parts, denoted Ω_1 and Ω_2 . Note that in this case there is no assumption on whether M has boundary or not.

Corollary 3 (Cheeger). *Let M be a compact Riemannian manifold. If $\partial M \neq \emptyset$, let $\mu_1(M)$ be the first Dirichlet eigenvalue on M and $\lambda_1(M)$ be its first nonzero Neumann eigenvalue for the Laplacian. When $\partial M = \emptyset$, we will denote the first nonzero eigenvalue of M by $\lambda_1(M)$ also. Then*

$$\mu_1(M) \geq \frac{ID_1(M)^2}{4}$$

and

$$\lambda_1(M) \geq \frac{IN_1(M)^2}{4}.$$

Theorem 12. *Let M be a (compact) Riemannian manifold with boundary. The Dirichlet heat kernel is bounded from above by*

$$H(x, y, t) \leq \left(\frac{2C_{SD}}{m} \right)^{-m/2} t^{-m/2},$$

for all $x, y \in M$ and $t > 0$, where $C_{SD} > 0$ is the Sobolev constant from (10.5). Moreover, the k th Dirichlet eigenvalue must satisfy the estimate

$$\mu_k \geq \frac{2eC_{SD}}{m} \left(\frac{k}{V(M)} \right)^{2/m}$$

and the k th Dirichlet eigenfunction must satisfy

$$\|\phi_k\|_\infty^2 \leq e \left(\frac{2C_{SD}}{m} \right)^{-m/2} \mu_k^{m/2} \|\phi_k\|_2^2$$

Remark. In compact manifold, let us recall that Sobolev inequality asserts that for $m \geq 3$ there exists a constant $C_{SD} > 0$ depending only on M , such that

$$\int_M |\nabla f|^2 \geq C_{SD} \left(\int_M |f|^{2m/(m-2)} \right)^{(m-2)/m}$$

for all $f \in H_c^{1,2}(M)$. In fact, Corollary 9.8 implies that $C_{SD} \geq ((m-2)/(2m-2)ID_{m/(m-1)}(M))^2$. When $m = 2$, the Sobolev inequality can be taken to be

$$\int_M |\nabla f|^2 \geq C_{SD} \left(\int_M |f|^p \right)^{2/p}$$

for all $f \in H_c^{1,2}(M)$ and for any fixed $2 \leq p < \infty$. For our purpose, we do not need the precise value of the Sobolev constant C_{SD} , beyond the fact that it only depends on the manifold M .

Remark. There's a similar result for the k th Neumann eigenvalue.

Theorem 13. *Let M be an m -dimensional manifold with boundary ∂M . Assume that M is geodesically star-shaped with respect to a point $p \in M$. Suppose that the Ricci curvature of M is bounded from below by*

$$\mathcal{R}_{ij} \geq -(m-1)R$$

for some constant $R \geq 0$. Let ρ be the radius of the largest geodesic ball centered at p contained in M , and ρ_0 be the radius of the smallest geodesic ball centered at p containing M . Then there exists a constant $C_1 > 0$ depending only on m , such that the first nonzero Neumann eigenvalue $\lambda_1(M)$ has a lower bound given by

$$\lambda_1(M) \geq \frac{\rho^m}{\rho_0^{m+2}} \exp\left(-C_1\left(1 + \rho_0\sqrt{R}\right)\right).$$

Corollary 4. *Let $B_p(\rho)$ be a geodesic ball centered at a point $p \in M$ with radius $\rho > 0$ such that $B_p(\rho) \cap \partial M = \emptyset$. Suppose that the Ricci curvature of $B_p(\rho)$ is bounded from below by*

$$\mathcal{R}_{ij} \geq -(m-1)R$$

for some constant $R \geq 0$. Then there exists a constant $C_2 > 0$ depending only on m , such that the first nonzero Neumann eigenvalue $\lambda_1(B_p(\rho))$ has a lower bound given by

$$\lambda_1(B_p(\rho)) \geq \rho^{-2} \exp\left(-C_2(1 + \rho\sqrt{R})\right).$$

Theorem 14. *Let M be a compact Riemannian manifold with smooth boundary ∂M . If there exists a positive function f defined on M satisfying*

$$\Delta f \leq -\mu f,$$

then the first Dirichlet eigenvalue, $\mu_1(M)$, of M must satisfy

$$\mu_1(M) \geq \mu.$$

In particular, if M is complete, noncompact, and without boundary, and the positive function f described above is defined on M , then $\mu_1(M) \geq \mu$.

2.1 Non-compact case

When M is compact, it's well known that $\sigma(\Delta)$ is discrete and $L^2(M)$ has a decomposition wrt eigenfunctions. There has been much work done on the estimates of eigenvalues, especially the first one, since it's the largest positive constant in Poincaré's inequality. When M is non-compact, not much is known [4].

One defines the **essential spectrum** $\sigma_{\text{ess}}(-\Delta)$ to be those λ in the spectrum which are either accumulation points of the spectrum or eigenvalues of infinite multiplicity.

It is well-known that if M is an n -dimensional simply connected complete manifold with constant curvature $-c \leq 0$, then its essential spectrum coincides with the spectrum, being such the interval $[(n-1)^2 c/4, \infty)$.

Theorem 15 (H. Donnelly and P. Li). *Let M be a complete Riemannian manifold and suppose $\bar{K}(r) \rightarrow -\infty$ as $r \rightarrow \infty$. Then the essential spectrum of Δ is empty provided one of the following two side conditions is satisfied: (i) M is simply connected and negatively curved. (ii) M is two dimensional and the fundamental group of M is finitely generated.*

Remark. Fix a point $p \in M$ and write $\bar{K}(r) = \sup\{K(x, \pi) \mid d(p, x) \geq r\}$ where $K(x, \pi)$ is the sectional curvature of a two plane π in $T_x M$.

Theorem 16 ([3]). *Let M be a non-compact complete Riemannian manifold whose Ricci curvature is bounded below by $-(d-1)c, c \geq 0$. Then the essential spectrum of Δ intersects the interval $[0, (d-1)^2 c/4]$.*

3 Spectral analysis of unbounded operators

Example 3. We denote by $L^2(\mathbb{R}^m)$ the space of (or class of) measurable functions on $\mathbb{R}^m$ that are square integrable with respect to the Lebesgue measure dx (for $dx_1, \dots, dx_m$). The Laplacian

$$-\Delta = -\sum_{j=1}^m \frac{\partial^2}{\partial x_j^2}$$

has no eigenfunctions in $L^2(\mathbb{R}^m)$, i.e., there is not a pair (u, λ) with $\lambda \in \mathbb{C}$ and $u \neq 0$ in L^2 such that $-\Delta u = \lambda u$ in the sense of distributions. But it has, for any $\lambda \in \mathbb{R}^+$, an eigenfunction in $\mathcal{S}'(\mathbb{R}^m)$ (the space of tempered distributions) (actually, in $L^\infty(\mathbb{R}^m)$) and, for any $\lambda \in \mathbb{C}$, an eigenfunction in $\mathcal{D}'(\mathbb{R}^m)$ (the space of distributions).

Example 4. The differential operator $-\Delta$ with domain $\mathcal{H}_0 = C_0^\infty(\mathbb{R}^m)$ is essentially self-adjoint. Its self-adjoint extension is $-\Delta$ with domain H^2 .

Example 5. The differential operator $-\Delta$ with domain $C_0^\infty(\Omega)$ (where Ω is an open bounded set with a smooth boundary) is not essentially self-adjoint. There exist many self-adjoint extensions related to the choice of boundary problems. As we have seen before, we have already met two such extensions: - the Dirichlet realization, whose domain is the set $H_0^1(\Omega) \cap H^2(\Omega)$; - the Neumann realization, whose domain is the set

$$\{u \in H^2(\Omega) \mid (\partial u / \partial \nu)_{/\partial \Omega} = 0\}.$$

Example 6. The Laplace-Beltrami operator on a compact manifold M with domain $C^\infty(M)$ is essentially self-adjoint on $L^2(M)$. The domain of the self-adjoint extension is $H^2(M)$ (which can be described using local charts as the space of the distributions that are locally in H^2).

Theorem 17. Let T be a self-adjoint operator. Then $\lambda \in \sigma(T)$ if and only if there exists a sequence $(u_n)_{n \in \mathbb{N}}$, $u_n \in D(T)$, such that $\|u_n\| = 1$ and $\|(T - \lambda)u_n\| \rightarrow 0$ as $n \rightarrow +\infty$.

Definition 4. A symmetric operator T with domain $\mathcal{H}_0$ is called **essentially self-adjoint** if its closure is self-adjoint.

Proposition 1. If T is essentially self-adjoint, then its self-adjoint extension is unique.

Proposition 2. Let T be a self-adjoint operator. A real λ is in the **discrete spectrum** if and only if λ is an isolated point in $\sigma(T)$ and an eigenvalue of finite multiplicity (that is, with a corresponding eigenspace of finite dimension).

Definition 5. The **essential spectrum** is the complement in the spectrum of the discrete spectrum. a point in the essential spectrum corresponds to either:
a point in the continuous spectrum;
a limit point of a sequence of eigenvalues with finite multiplicity; or
an eigenvalue of infinite multiplicity.

Proposition 3. The essential spectrum of a self-adjoint operator T is closed in $\mathbb{R}$.

Example 7. The essential spectrum of a compact self-adjoint operator is reduced to 0.

The essential spectrum of an operator with a compact resolvent is empty.

The Laplacian on $\mathbb{R}^m$ is a self-adjoint operator on $L^2(\mathbb{R}^m)$ whose domain is the Sobolev space $H^2(\mathbb{R}^m)$. The spectrum is continuous and equal to $\overline{\mathbb{R}^+}$. The essential spectrum is also $\overline{\mathbb{R}^+}$, and the operator has no discrete spectrum.

Theorem 18 (Weyl sequence). Let T be a self-adjoint operator. Then λ belongs to the essential spectrum if and only if there exists a sequence u_n in $D(T)$ with $\|u_n\| = 1$ such that u_n tends weakly to 0 and $\|(T - \lambda)u_n\| \rightarrow 0$ as $n \rightarrow \infty$.

Theorem 19. *Let T be a self-adjoint operator, and let V be symmetric and T -compact. Then $T + V$ is self-adjoint, and the essential spectrum of $T + V$ is equal to the essential spectrum of T .*

Proposition 4. *Let A be a self-adjoint operator. Then $A \geq 0$ if and only if $\sigma(A) \subset [0, +\infty[$.*

Theorem 20. *Let A be a self-adjoint semibounded operator. Let $\Sigma := \inf \sigma_{\text{ess}}(A)$, and consider $\sigma(A) \cap]-\infty, \Sigma[$, described as a sequence (finite or infinite) of eigenvalues $\lambda_j(A)$ (or, more simply, λ_j), which is ordered in increasing order:*

$$\lambda_1 < \lambda_2 < \cdots < \lambda_n \cdots$$

Then we have

$$\begin{aligned} \lambda_1 &= \inf_{\phi \in D(A), \phi \neq 0} \|\phi\|^{-2} \langle A\phi, \phi \rangle, \\ \lambda_2 &= \inf_{\phi \in D(A) \cap K_1^\perp, \phi \neq 0} \|\phi\|^{-2} \langle A\phi, \phi \rangle, \end{aligned}$$

and, for $n \geq 2$,

$$\lambda_n = \inf_{\phi \in D(A) \cap K_{n-1}^\perp, \phi \neq 0} \|\phi\|^{-2} \langle A\phi, \phi \rangle$$

where

$$K_j = \oplus_{i \leq j} N(A - \lambda_i)$$

Theorem 21. *Let M be a non-compact complete Riemannian manifold whose Ricci curvature is bounded below by $-(d-1)c, c \geq 0$. Then the essential spectrum of Δ intersects the interval $[0, (d-1)^2 c/4]$.*

A somewhat more interesting invariant of M , which we denote by λ_0^{ess} , is the greatest lower bound of the essential spectrum of Δ , where the essential spectrum consists of points of the spectrum of Δ which are either accumulation points of points on the spectrum or which correspond to discrete eigenvalues of Δ with infinite multiplicity. It is classical that if M is compact, the essential spectrum is empty. We also clearly have that $\lambda_0 \leq \lambda_0^{\text{ess}}$, and that

$$\lambda_0^{\text{ess}}(M) = \lim_K \lambda_0(M - K), \quad (8)$$

where K runs through all compact subsets of $M[1]$.

Remark. See also [5], theorem 11.18.

Proposition 5 (cf. F. Haslinger's note, Spectral analysis). *Let T be an unbounded self-adjoint operator on H which is non-negative in the sense that $\sigma(T) \subseteq [0, \infty)$. Then the following conditions are equivalent:*

- (i) *The resolvent operator $(I + T)^{-1}$ is compact.*
- (ii) *$\sigma_e(T) = \emptyset$.*
- (iii) *There exists a complete orthonormal system of eigenvectors $\{x_n\}_n$ of T with corresponding eigenvalues $\mu_n \geq 0$ which converge to $+\infty$ as $n \rightarrow \infty$.*

Remark. In the paper [6], they find the following properties:

Let H_1, H_2 be Hilbert spaces and T be a closed linear operator defined on a dense subspace $D(T)$ in H_1 and taking values in H_2 . In this article we prove the following results:

(i) Range of T is closed if and only if 0 is not an accumulation point of the spectrum $\sigma(T^*T)$ of T^*T ,

In addition, if $H_1 = H_2$ and T is self-adjoint, then (ii) $\inf \{\|Tx\| : x \in D(T) \cap N(T)^\perp, \|x\| = 1\} = \inf \{|\lambda| : 0 \neq \lambda \in \sigma(T)\}$,

(iii) Every isolated spectral value of T is an eigenvalue of T ,

(iv) Range of T is closed if and only if 0 is not an accumulation point of the spectrum $\sigma(T)$ of T ,

(v) $\sigma(T)$ bounded implies T is bounded.

References

- [1] Robert Brooks. A relation between growth and the spectrum of the laplacian. *Mathematische Zeitschrift*, 178(4):501–508, 1981.
- [2] Shiu-Yuen Cheng. Eigenvalue comparison theorems and its geometric applications. *Math. Z.*, 143:289–297, 1975.
- [3] Harold Donnelly. On the essential spectrum of a complete riemannian manifold. *Topology*, 20(1):1–14, 1981.
- [4] JoséF Escobar. On the spectrum of the laplacian on complete riemannian manifolds. *Communications in partial differential equations*, 11(1):63–85, 1986.
- [5] Bernard Helffer. *Spectral theory and its applications*, volume 139 of *Camb. Stud. Adv. Math.* Cambridge: Cambridge University Press, 2013.
- [6] Sudhir H Kulkarni, M Thamban Nair, and Golla Ramesh. Some properties of unbounded operators with closed range. *Proceedings Mathematical Sciences*, 118:613–625, 2008.
- [7] Shing-Tung Yau. Some function-theoretic properties of complete Riemannian manifold and their applications to geometry. *Indiana Univ. Math. J.*, 25:659–670, 1976.